



Techniques for Mitigating the Vanishing Gradient Problem in Deep Neural Networks

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Abstract. The Vanishing Gradient (VG) problem is a significant challenge during Deep Neural Networks (DNN) training, mainly when using the backpropagation algorithm. This issue is especially true for tasks involving sequential data and long-range dependencies, such as natural language processing (NLP). Over the years, the VG problem has received significant research attention, indicating its importance. However, the few existing studies on the VG problem that offer a subset of work in this area do not provide a structured understanding of the various methods to solve the VG problem. This study briefly explores methods proposed to mitigate the VG problem, categorizing them into three main approaches: (1) modified activation functions, such as ReLU and its variants, which help alleviate gradient shrinkage in deep layers; (2) architectural innovations, such as residual connections and LSTMs, designed to preserve gradient flow over long sequences, and (3) optimization strategies, including advanced methods like Adam, which adapt learning rate dynamically to stabilize training. A comparative analysis of these techniques highlights key insights: while ReLU variants provide simplicity and computational efficiency, they may still struggle with dying neurons; architectural approaches like residual networks offer robust gradient propagation and great potential but at the cost of increased complexity—optimization methods like Adam's balance computational overhead with improved convergence rates. We conclude by discussing potential future directions for further research in mitigating the vanishing gradient problem, instilling a sense of optimism for the future of deep learning.

Keywords: Vanishing gradient · deep neural network · backpropagation algorithm

1 Introduction

Deep Neural Networks (DNN) have transformed domains such as computer vision and text generation by enhancing the modelling, and representations of complex patterns. However, training these networks often encounters a significant obstacle known as the

vanishing gradient problem [1]. This problem, a consequence of the fundamental back-propagation algorithm in training DNN, poses a critical challenge, particularly for networks dealing with high-dimensional data such as texts. Figure 1 illustrates a DNN with multiple layers of input, hidden and output layers, which shows the flow of gradients through the network.

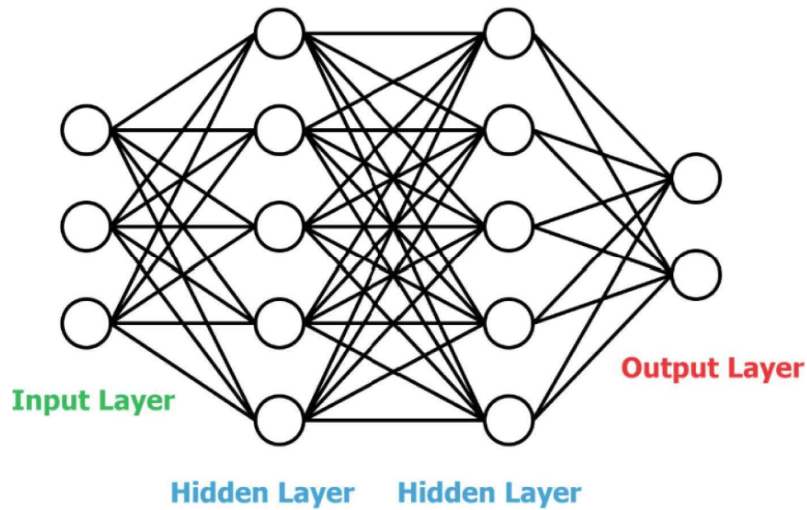


Fig. 1. Illustration of the DNN architecture

As neural networks deepen, the gradients used for updating the weights during back-propagation can become increasingly smaller or completely vanish, limiting the learning process and making it difficult for the model to preserve long-term relationships within the data [2].

The vanishing gradient is not just applicable to DNN but a particularly pronounced one in NLP with tasks such as language modelling, machine translation, and text generation, which necessitate the learning and representing intricate relationships and dependencies within textual data. The sequential nature of language and the prevalence of long-range dependencies make NLP tasks especially vulnerable to the VG problem, adding another layer of complexity to an already challenging issue [3]. However, limited research has been conducted on the VG problem. Hence, this survey is generally about VG in various DNN applications.

The vanishing gradient refers to the phenomenon where the gradients for the parameters of a neural network become very small as more layers are added, using certain activation functions like sigmoid or traditional ReLU. This can cause the network weights to be updated very slowly and ineffectively, leading to difficulty learning complex patterns in text data. The VG problem has been recognized as an issue since the early days of DNN [4]. Moreover, it has hindered the efficient training of deep architectures.

Over the years, researchers have proposed various methods to mitigate the VG problem. One approach has been to modify the activation functions, such as introducing variants of activation functions such as Leaky ReLU and parametric ReLU, which help maintain non-zero gradients even in deeper layers [1]. However, these modifications do

not provide a complete solution, and their effectiveness varies across different tasks and datasets.

Another method explored is architectural innovations like residual networks and Transformer models, developed to preserve gradient flow across layers, enabling the model to learn complex long-range dependencies effectively [5]. Optimization strategies like gradient clipping and Adam optimization, further stabilize the training process by preventing the gradients from vanishing or exploding.

Despite these efforts, the VG problem remains a significant challenge, particularly in text data and NLP tasks, where deep neural networks must effectively learn and represent long-term dependencies and complex relationships.

Several papers have recently been published on the VG problem. However, few literature surveys are available to review it. [6] provides an overview of activation functions in DNN. The author explicitly introduces activation functions and how they are used in DNN; their survey is limited to modifying activation functions to solve the VG problem. [7, 8] similarly provide a good understanding of the different types and characteristics of activation functions across various DNN applications. However, they are limited to modifying activation functions to solve the VG problem. To the best of our knowledge, no survey captures all methods used to solve the VG problem. Therefore, this current review aims to fill the gap by investigating the VG problem and the various methods researchers use to mitigate it. The following sections of this paper will investigate the details of various methods, such as modification of activation functions, architectural innovation, and optimization methods, used to mitigate the VG problem. This is followed by discussions and conclusions.

2 Vanishing Gradient Concept

Deep learning is an Artificial Neural Network (ANN) that requires multiple layers of information processing stages in hierarchical architectures. These layers are trained using the backpropagation network algorithm, a widely used technique for training deep neural networks to classify and identify patterns [9]. The backpropagation algorithm is designed to modify the network weights by making incremental adjustments in the opposite direction of the error derivative. In deep networks, however, the gradients become increasingly smaller for each layer further from the output, a phenomenon known as the vanishing gradient problem [10]. The VG problem can be understood in a situation where the gradient of the objective function to the parameter approaches zero, causing minimal or no parameter updates during the network training, which leads to poor convergence and eventually stops training [7]. A significant cause of this problem is the choice of activation function [1]; a mathematical function is applied to each neuron's output in a neural network to determine whether or not the neuron should be activated. The activation function is applied in every neuron of the DNN to learn complex data patterns for accurate network predictions [1]. In the course of training Deep Neural Networks (DNNs), the weights and biases for each layer are determined by calculating the discrepancy between the predicted output and the actual label. However, in the beginning layers of the network, weights and bias updates may gradually become small due to activation functions such as sigmoid, which can lead to the VG problem

and eventually stop the learning process. Figure 2 is a typical illustration of activation functions, such as the sigmoid function.

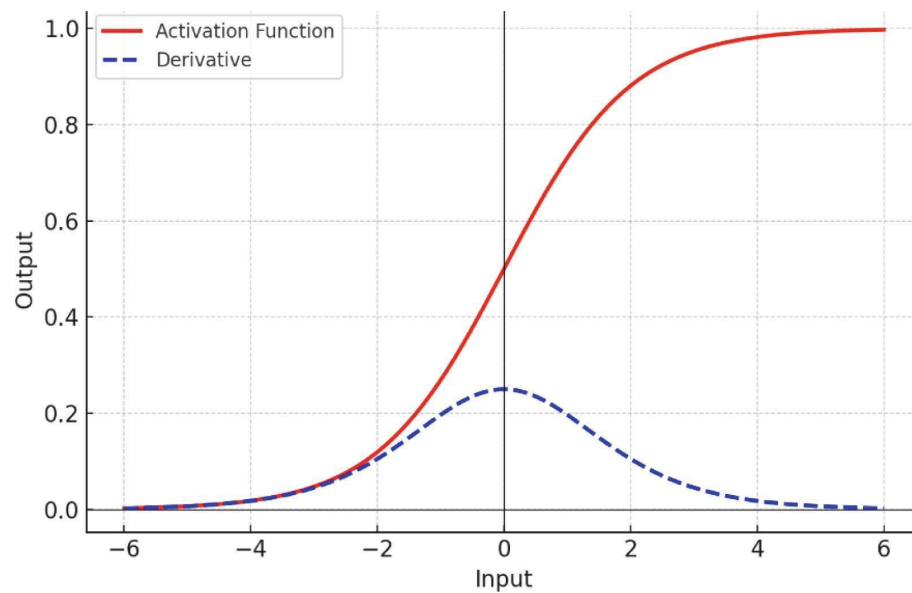


Fig. 2. Illustration of the activation function in DNN

This function compresses its input into a value between 0 and 1, which causes the output to approach 0 or 1 respectively. However, as the input becomes larger or smaller, the sigmoid function's derivative (represented by the blue curve) becomes very small. During backpropagation, gradients are calculated based on the derivative of the activation function. Since the derivative of the sigmoid function approaches zero, the gradient diminishes gradually, causing weight updates in earlier layers to become very small. This can prevent effective layer learning, slowing down the DNN training or causing it to stop completely, known as the VG problem.

As a result, DNNs such as Recurrent Neural Networks (RNNs) face this issue in their earlier layers, making it challenging to learn long-term dependencies effectively. This is because the gradients become too small to provide meaningful updates to the network weights, leading to slow or ineffective training [11].

In natural language processing (NLP) tasks, recurrent neural networks (RNNs) are commonly employed because of their capacity to examine words individually and consider the preceding state output as the current state input. However, RNNs are known to perform well only for a few text analyses and can face challenges when dealing with long-term dependencies [12].

Various modifications have been made to the activation functions and neural networks to address these issues. The following section will examine the methods employed to solve the VG problem.

3 Methods to Solve Vanishing Gradient

3.1 Modified Activation Functions

Modifying the ReLU and sigmoid activation functions is one of the most widely used methods to mitigate the VG problem. This is because these two are the commonly used activation functions for neural networks, and they are usually faced with VG problems. The underlying concept of sigmoid and ReLU is described by [13] with two crucial mathematical formulas. The first is the sigmoid activation function, which takes the form:

$$\text{Sigmoid}(x) = \frac{1}{1 + \exp(-x)} \quad (1)$$

the sigmoid function describes a smooth, S-shaped curve that interprets the outputs as a value probability between 0 and 1 for any real-valued input x . This characteristic allows it to convert a wide range of input values to a normalized output, making it useful for binary classification tasks in machine learning models. This compression can lead to a VG issue because its derivative is typically small and progressively smaller when it propagates through the multiple layers of the deep neural network. The other formula that describes the ReLU is in the form of:

$$\text{ReLU}(x) = \max(0, x) \quad (2)$$

where the ReLU function generates a constant output of 0 for any negative input. This can result in the dying ReLU phenomenon, where neurons output zero, and the gradients do not change, resulting in a VG problem variant.

Thus, changing the activation function through modification of the sigmoid [14] or reinstating the original derivative of the sigmoid to an artificial one for practical model training [13]. These modified activation functions have shown improved performance in handling VG issues.

3.2 Architectural Innovations

Recent architectural innovations have shown great potential for tackling deep neural networks' vanishing gradient (VG) issue. Unlike modifying activation functions, which focus on the individual neuron level, these architectural advancements introduce structural changes to the network architecture, facilitating more efficient gradient flow during the backpropagation process. Architectural innovations have demonstrated remarkable success in various deep-learning tasks, including natural language processing and sequential data modeling, enabling better gradient propagation.

[15] proposed a Recurrent Identity Network (RIN) to overcome the VG in the traditional recurrent network. The RIN model employs parallel computations and separates the influence of input and hidden states during the recurrent computation. This separation prevents the gradients from vanishing, as the gradients are computed independently for the input and hidden states. Crucially, the RIN architecture leverages the ReLU activation function, significantly speeding up the convergence and enabling the network to be less sensitive to the training conditions.

Another architectural innovation in RNNs is the Independently recurrent network termed (the IndRNN) model [16]. In this model, neurons are independent of each other, allowing for effective gradient propagation over different time steps and maintaining long-term memory. The IndRNN model has been successfully applied to language modelling and image classification tasks, demonstrating its ability to mitigate the VG problem. Also, BiGridLSTM architecture [5] has been proposed to alleviate the VG problem in Long Short-Term Memory (LSTM) networks. This architecture combines an LSTM grid and bidirectional architecture, enabling more effective gradient propagation and improving the performance of LSTM models on speech datasets.

The Transformer architecture [17] has effectively addressed the VG problem in tasks involving sequential data. Transformer-XL, with its self-attention mechanism, mitigates the VG problem by providing a mechanism to retain and utilize information from distant parts of the input sequence, leading to improved performance in capturing long-range dependencies in a variety of NLP tasks [18].

Therefore, architectural innovations have been crucial in mitigating the VG problem, particularly in tasks involving sequential data and long-range dependencies, such as natural language processing. By introducing structural changes that facilitate more efficient gradient flow, these architectures have significantly enhanced DNNs' representational capacity and performance in capturing complex patterns and relationships in sequential data.

3.3 Optimization

Optimization methods focus on efficiently updating the network's parameters during training and ensuring that the gradients are effectively utilized for learning. While architectural innovations and activation function modifications aim to facilitate adequate gradient flow, optimization methods have been widely used in conjunction with architectural innovations [19].

One optimization technique that has shown promise in addressing the VG issue is the Stochastic Diagonal Approximate Greatest Descent (SDAGD) algorithm. [1] demonstrated that SDAGD, when used with ReLu and sigmoid activation functions, can lead to faster training and better performance than standard stochastic gradient descent (SGD), which often struggles with slower convergence and VG problems.

Adaptive learning rate optimization algorithms, such as Adaptive Moment Estimation (Adam) and Root Mean Square Propagation (RMSProp) optimizer, have also been explored as effective methods for mitigating the VG problem and utilized the Adam optimizer in conjunction with recurrent neural network (RNN) models for text classification tasks, demonstrating its ability to stabilize the training process and overcome the VG issue [2]. Similarly, RMSProp resists the VG problem and has been successfully applied in tasks like fake news detection [20]. Another widely employed optimization method is gradient clipping [15], which involves constraining the gradients within a predetermined range to stabilize the optimization process and reduce the impact of the VG issue.

In addition to these techniques, normalization and regularisation methods have also been explored. Batch normalization and regularization techniques like dropout have been utilized in text classification tasks to help mitigate the VG issue [3].

Recent studies have shown that optimization methods can be enhanced by incorporating architectural innovations and adjusting activation functions. This can improve performance in deep neural network models and streamlined training processes, ultimately addressing the VG problem, particularly in natural language processing tasks.

4 Discussion

4.1 Modified Activation Functions

The first advantage of this method is that it requires less computational cost. Another advantage is that it makes little changes to the neural network. While these modifications have shown promising results, their effectiveness can vary across tasks and datasets. Moreover, the choice of activation function modification often needs to be customized to fit the specific needs and traits of the problem regarding aspects such as the network structure, optimization strategy, and the nature of the data. As observed from this review, studies here focus on activation functions, and the majority were tested on image datasets [13, 14]. As researchers have pointed out, this method's disadvantage is domain-specific and may lead to training instability.

4.2 Architectural Innovations

The first advantage of architectural innovation is that it improves model performance [15]. Research shows that most of the architectural innovation in DNN is focused on RNN due to its wide usage in high-dimensional data. However, it cannot memorize long-term information, leading to VG problems. Another advantage of this method is that it facilitates efficient gradient flow because it allows for structural changes to network architecture. Despite this method's flexibility to improve neural networks, it requires high computational costs and increased network complexity. However, techniques such as the transformer model prove more effective by working independently without additional objectives [18], resulting in faster training and mitigating vanishing gradient problems.

4.3 Optimization

One advantage of the optimization method is its flexibility in training neural networks and its ability to learn complex patterns and long-range dependencies in data, such as NLP tasks [15]. However, they can introduce additional hyperparameter tuning, which is time-consuming and requires high computational cost. Also, other approaches in this method, such as batch normalization, may not be practical for fewer datasets or tasks where the input distribution frequently changes. In general, while optimization can efficiently reduce the VG problem, it is essential to carefully select the most appropriate technique based on specific datasets and tasks.

Specifically, VG issues are more pronounced in NLP tasks due to their sequential nature, which limits the ability of architectures like RNNs to capture long-sequence tasks such as language modelling.

Table 1 provides a concise overview of the advantages and disadvantages of the previously mentioned methods.

Table 1. Summary of advantages and disadvantages of various methods for addressing the VG problem.

Method	Pros	Cons
Modified activation functions	-Requires less computational cost - Ensures faster convergence -Can be integrated easily with little changes into an existing neural network	-It is domain-specific -Leads to training instability
Architectural innovations	- Facilitate efficient gradient flow -Improves model performance	-Effectiveness varies across tasks and datasets -Requires high computational cost
Optimization	-Flexible for hyperparameter tuning -Able to learn long-range dependencies	-Time consuming -Requires high computational cost

5 Further Research Directions

Despite substantial progress in mitigating the VG problem, critical gaps that require further research remain. While effective in some domains, current methods often fail to generalize across different tasks, mainly in NLP and other applications requiring the modelling of long-term dependencies. Future research should focus on developing more adaptive, scalable, and task-specific solutions to address this.

1. Task-Specific and Dynamic Activation Functions:

Current activation functions, such as ReLU and its variants, address the VG problem to a degree but often lack adaptability for specific tasks, especially those involving sequential or highly nonlinear data. A promising research direction would be creating task-specific activation functions that adapt dynamically during training. These functions could vary their behaviour based on the depth of the network, the layer context, or the characteristics of the input data. These activation functions could ensure better gradient propagation by incorporating mechanisms that adjust the gradient behavior in real-time, particularly in tasks like NLP and time-series analysis. Dynamic activation functions that evolve during training could mitigate VG in deeper networks, offering a more nuanced solution than static functions like Leaky ReLU.

2. Hybrid Architectures and Optimization Techniques:

While residual connections and Transformer-based architectures have shown promise in addressing gradient flow, there is still significant untapped potential in combining architectural innovations with adaptive optimization strategies. For instance, hybrid models that integrate self-attention mechanisms with recurrent architectures could provide short-term and long-term dependency modeling, a crucial requirement for speech recognition and reinforcement learning tasks. These hybrid architectures could also incorporate dynamic optimizers, like an enhanced version of Adam, which

adjusts learning rates and model depth or connectivity in response to gradient behaviors. Optimization strategies, such as layer-wise gradient clipping and advanced normalization techniques, could prevent vanishing gradients while maintaining efficient training.

3. Regularization Techniques Tailored to Gradient Behavior:

Often employed to prevent overfitting, regularisation methods could also be leveraged to stabilize gradient propagation. Gradient-sensitive regularization techniques, such as layer-wise regularization penalties that adjust dynamically based on gradient magnitude, could help mitigate vanishing and exploding gradients. By selectively applying more robust regularization in deeper layers or where gradients are more likely to vanish, such methods could improve training stability in very deep networks. Techniques like variational dropout or adaptive weight regularization might also be re-imagined to directly counteract VG by enforcing constraints that maintain healthy gradient flow throughout the network.

4. Transfer Learning and Pretrained Models in VG Mitigation:

Using transfer learning, pre-trained models, such as BERT and GPT, can circumvent some of the difficulties associated with VG. However, the role of pre-trained models in VG mitigation has not been thoroughly investigated. Future research should explore how transfer learning strategies can be optimized to alleviate VG, particularly in low-resource scenarios. Additionally, there is room to explore the fine-tuning dynamics of these models, focusing on how gradient behavior changes across different layers during fine-tuning for specific tasks. Understanding how transfer learning interacts with VG-related challenges could inform new model initialization and training techniques, making it possible to train deeper models without encountering significant gradient decay.

5. Dynamic Neural Network Architectures:

One of the most promising future directions is the exploration of dynamic neural network architectures that adapt their depth, learning rates, or even internal structure based on gradient behavior during training. Dynamic architectures could adjust layer connections or neuron activation in real-time, reducing the likelihood of gradient vanishing in specific network parts. For example, layer-skipping mechanisms could be activated when gradients are too small, while specific layers could be expanded when gradients are too large. These architectures would evolve during training, making them more resilient to the VG problem.

6 Conclusion

This paper presented a brief review of the methods to handle VG, which can be utilized to solve the VG problem. We investigated the overview of the VG problem, its underlying causes, and various methods proposed to mitigate it. As we examined each method, we found that the modified activation functions, a popular approach to solving the VG problem, do not seem suitable for tasks with long-range dependencies, such as NLP tasks. Architectural innovations seem to be the best method, but their effectiveness has not been widely explored in literature. The optimization method looks promising. However, it is time-consuming and requires high computational requirements. Accordingly, these are some of the methods utilized to solve the VG problem. However, careful consideration

is required for NLP tasks, which require a more robust method due to the nature of text data. As a result, given the current literature on VG, new methods have been researched for the VG problem. Some of these methods utilize techniques such as the Transformer model, as observed in [18]. However, these methods have not been widely explored on various tasks, such as in the context of text data. Therefore, further innovations and research are needed to address this issue, as there seems to be no specific method fit for all data types.

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